

PRATHAM SONI

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Professional Summary

Master of Data Science and Decisions candidate at UNSW (Business Data Science concentration) working across the full analytics-to-AI spectrum: business intelligence and dashboarding on one end, and end-to-end machine learning pipelines on the other. Fluent in Python, SQL, and Power BI (DAX), with applied machine learning across ensemble methods, deep learning, NLP, computer vision, and time-series forecasting, hands-on model-serving experience (FastAPI, OAuth 2.0), three IEEE-published papers, and a first-prize international hackathon win. Comfortable owning a problem from messy raw data through modelling and evaluation to a decision a stakeholder can act on. Seeking graduate Data Analyst, Business Analyst, Data Scientist, or AI/ML Engineer roles in Sydney.

Technical Skills

Programming & Query: Python (pandas, NumPy, scikit-learn, XGBoost, TensorFlow/Keras, Seaborn, Matplotlib), SQL, Excel (advanced, Power Query, pivot/lookup).

Business Intelligence & Visualisation: Power BI (DAX, star-schema modelling), Tableau, dashboard design, data storytelling, exploratory data analysis (EDA).

Machine Learning: Regression & Classification, Ensemble Methods (XGBoost, Random Forest), Time-Series Forecasting (STL decomposition), Clustering (K-Means, DBSCAN, PCA), Cross-Validation, Feature Engineering & Ablation, Imbalance Handling (SMOTE, focal loss), Model Interpretability (feature importance).

Deep Learning & AI: CNN, RNN, LSTM, MLP; NLP (BERT, XLNet, GloVe, FastText, transformers); Computer Vision (YOLOv8, VGG16); Recommender Systems.

Data Engineering & Deployment: FastAPI, Flask, REST APIs, OAuth 2.0, model serving, Power Automate, web scraping, ETL / data cleaning, scheduled pipelines, Git.

Testing & Quality: Exploratory testing, negative-path and edge-case testing, cross-device and cross-browser testing, usability and user-journey review, defect logging and resolution, user acceptance walkthroughs and client sign-off.

Databases: SQL, PostgreSQL, MySQL, MongoDB.

Ways of Working: Agile/Scrum, stakeholder communication, data governance & quality, requirements framing.

Machine Learning Projects

LLM Agent Pipeline for Building Time-Series Classification | UNSW Master's Capstone (DATA5925)

- Adapted ZARA, a training-free evidence-grounded LLM agent framework built for human activity recognition, to Brick schema building-automation metadata, testing whether zero-shot agentic reasoning transfers across time-series domains without task-specific training.
- Built the pipeline end to end: 496 engineered features per daily signal, a 4,095-entry pairwise XGBoost feature-importance knowledge base across 91 composite classes, class-conditional FAISS retrieval over Mantis-8M embeddings, and a four-agent LLM decision chain (Feature Selector, Candidate Pruner, Refiner, Adjudicator) with schema-constrained JSON validation.
- Evaluated on a held-out partition of the official Brick-by-Brick 2024 test set: retrieval placed the true class in the top ten for 86.0% of queries, while the full pipeline reached 36.0% accuracy (macro F1 0.395) against 57.0% (macro F1 0.500) for an independently reproduced and verified run of the competition's winning CatBoost ensemble.
- Localised the shortfall to the reasoning stage rather than retrieval through controlled ablations: disabling candidate pruning left accuracy unchanged, and a rank-anchoring test showed the final agent defaulted to the top-retrieved candidate 26 to 47 points above chance, overriding a misleading rank in only 15 to 48% of cases.
- Demonstrated that the trained model and the zero-shot pipeline converge independently on the same Status/Alarm/Command confusion cluster, evidence that the limitation lies in Brick's functional-role labels rather than in either classification paradigm.

Beer Portfolio Optimisation | UNSW Sydney (INFS5700) | Power BI, DAX, Python

- Analysed a US national beer distributor's stalled business (150 products, 7 regions, 5 sales channels, -0.4% revenue growth over two years) to answer a single brief: how to optimise the portfolio.
- Ran exploratory analysis in Python first (distributions, YoY growth by family and region, BCG classification), stress-testing every claim against the raw data before building any visual.
- Built a star-schema data model in Power BI over 11,000 transactions, authoring DAX measures for revenue, profit, margin, YoY growth, discount impact, and BCG class.
- Surfaced the headline insight (zero overlap between the top 10 products by units sold and the top 10 by profit margin), showing the company was investing in busy products rather than profitable ones.
- Traced 64% of all discount spend to the lowest-margin channel (Supermarket at 37.3% vs Bar/Restaurant at 45.1%) and quantified an \$80,234 profit-improvement opportunity from reallocating discount toward higher-margin channels.
- Mapped shifting regional demand (Wheat +100% in the Mountain region, IPA +26.8% on the West Coast, the Southeast rotating from Ale -26.1% to IPA +13.4%) and flagged that six of ten families were shrinking.
- Delivered a prescriptive playbook (promote 14 product-region-channel combinations, reposition 3 products for a +7.9% to +10.7% margin gain, phase out 7 underperformers, redirect freed revenue to Stars for an added \$9,920 profit) as a five-page descriptive, diagnostic, predictive, and prescriptive narrative.

- Documented limitations explicitly (no demand-elasticity data, so \$80,234 is a theoretical upper bound; a review-to-revenue correlation of $r=0.60$ is not causation), treating caveats as analytical maturity rather than weakness.

Coral Bleaching Severity Prediction | *UNSW Sydney (MATH3856)* | *Python, scikit-learn, XGBoost, TensorFlow/Keras*

- Built an end-to-end multiclass classification pipeline predicting coral-bleaching severity (four ordinal classes) on the harmonised BCO-DMO global reef dataset spanning seven monitoring programs, 1980–2020.
- Engineered a rigorous cleaning process, dropping columns with over 50% missing data, removing rather than imputing incomplete rows to protect integrity, and eliminating 11,000 exact duplicates (about 37% of records) to land a clean 21,418-observation, 20-feature dataset.
- Maintained two parallel encodings (label encoding for tree models, one-hot for neural networks, expanding to roughly 132 features) to match each model family's representational assumptions.
- Trained and compared five architectures (XGBoost, Random Forest, Logistic Regression, and shallow and deep MLPs) under stratified k-fold cross-validation with up to 100 runs per configuration, scaling features within each fold to prevent data leakage.
- Handled class imbalance methodically, applying SMOTE inside the training split only (tested with and without to isolate its effect) for tree models and focal loss for neural nets, and quantifying algorithm-specific effects (SMOTE degraded Logistic Regression by 12.6%; focal loss cut MLP accuracy by up to 16%).
- Identified XGBoost with SMOTE as the top model (macro PR-AUC 0.736, ROC-AUC 0.927; 0.873 PR-AUC on the actionable early-warning class), with Random Forest a near-equal second, showing ensemble averaging alone handles moderate imbalance.
- Ran ablation studies proving reef-structural features (exposure, depth, turbidity) contributed roughly 70% of the performance gain over temperature-only models, a result with direct scientific and operational meaning.
- Designed an operational deployment workflow (weekly sensor-CSV ingestion, XGBoost scoring on a web server emitting a binary risk flag, field-validation feedback, and periodic retraining) and justified XGBoost for production on speed, feature-importance interpretability, and reliability.

ATLAS: Retail Basket and Trend Dashboard | *UNSW DATA5002, group project*

- Prepared and analysed Australian Bureau of Statistics Retail Trade data (ABS 8501.0) for a multi-tab Streamlit dashboard linking micro-level basket behaviour to macro-level retail trends, alongside the UCI Online Retail II transactional dataset.
- Caught and corrected a turnover scaling error flagged in review by systematically cross-verifying calculated figures against official ABS releases and published economic reports, restoring confidence in monthly trends, seasonal peaks and year-on-year growth rates.
- Produced monthly turnover trends, seasonal decomposition and industry comparison views across six ABS retail industries, and documented the granularity mismatch between product-level transactions and industry-level aggregates explicitly in the dashboard.
- Designed against two stated user personas, a retail analyst and a store manager, balancing analytical depth against readability for non-technical users. Live at atlas-retail.streamlit.app.

Air Quality Forecasting: Hybrid STL Ensemble | *IEEE Published* | *Python, STL, CNN-LSTM, RNN*

- Forecast AQI, PM2.5, and PM10 across six Mumbai monitoring stations (January 2020 to May 2023) using a hybrid ensemble that decomposes the signal before predicting it.
- Split each series via STL into trend, seasonal, and residual components, then matched a specialist learner to each: a CNN-LSTM stack for trend, and stacked RNNs with dense heads for seasonal and residual dynamics, recombined by ensemble aggregation.
- Beat VAR, VARMA, and CEEMDAN baselines on every target, reaching R^2 0.89 on AQI (RMSE 13.68), 0.86 on PM2.5, and 0.75 on PM10, validated on RMSE, MAPE, and R^2 .
- Framed the work as a decision-support tool for policymakers (school-day advisories, traffic nudges, construction-shutdown thresholds), not a leaderboard exercise.

Disaster Tweet Classification with Deep Learning | *IEEE Published* | *BERT, XLNet, GloVe, FastText, LSTM*

- Classified 53,531 CrisisNLP tweets to separate genuine eyewitness reports from sarcasm, news commentary, and metaphorical noise that keyword filters miss.
- Benchmarked four embeddings (BERT, XLNet, GloVe, FastText) across feed-forward, CNN, and LSTM architectures, then built a hybrid ensemble (BERT + FastText + GloVe with an LSTM head) that averaged predictions to reach 89.12% accuracy, beating every individual model (82 to 88%).
- Found BERT strongest on context and FastText strongest on out-of-vocabulary and morphological variation, with balanced precision, recall, and F1 tuned for emergency triage.

Clear Waters: Plastic Detection in Rivers | *1st Prize, REVA and Kyndryl Hackathon* | *YOLOv8, FastAPI, Flutter*

- Led the end-to-end build of a drone-mobile-web platform that detects, maps, and forecasts river plastic, winning first prize against 750 teams from 19 countries.
- Trained a dual-YOLOv8 detector, a domain-specialist model (YOLOv8x) on 2,800 Saigon-River drone frames and a generalist (YOLOv8n) on 6,523 diverse images, merging their outputs via DBSCAN to keep the highest-confidence detections (50.3% mAP50-95 across 7+ trained variants).
- Curated the generalist's training set down from 25,000 noisy images to 6,523 using a VGG16, PCA, and K-Means clustering pipeline (40 clusters), which bought more accuracy than any architecture change.

- Built the serving layer in FastAPI with Dropbox storage over OAuth 2.0, handling single uploads and scheduled bulk drone batches, and auto-routing images to the right preprocessing pipeline from their metadata.
- Added a drift-prediction map (wind speed and direction, tide height, wave period) that snaps forecasts to the nearest water body, plus a Flutter field app and a Power BI dashboard for detection trends.

Adaptive Learning & Curriculum Recommender | *Final-Year Project, NMIMS | 2nd Place, SAAVISHKAR 2024*

- Built a recommender-driven EdTech platform that scores a student's prior knowledge with a pre-quiz, ranks modules by predicted gain and difficulty fit, and re-runs after every quiz so the curriculum adapts in-flight rather than per term.
- Turned the same model into a teacher-facing view, aggregating thousands of micro-interactions into a single per-class heatmap with material-allocation controls readable in seconds.
- Placed 2nd at SAAVISHKAR 2024, the NMIMS Computer Engineering final-year project competition.

Silver Oaks Luxury: E-commerce Store | *Freelance Full-Stack Developer | PHP, MySQL, HTML/CSS, JavaScript*

- Built a full-stack e-commerce store for a BIS-hallmarked luxury silver brand: a mobile-first catalogue across five collections with listing, detail, cart, checkout, and UPI order confirmation.
- Delivered a secure self-service admin panel (product, brand, order, and customer management with image uploads, manual tracking, order history, and CMS controls) so the client runs the store without developer involvement.
- Ran exploratory and negative-path testing across the checkout and payment flow, including interrupting transactions mid-payment to confirm the platform did not record an order it had not received, catching defects a happy-path walkthrough would have missed.
- Tested the storefront across devices and browsers for responsiveness and layout integrity, and walked the build as a first-time user to check attention landed on the actions the business wanted.
- Ran a live user acceptance walkthrough with the client across the storefront and admin panel, secured sign-off, and handed over a platform they operate independently.
- Scoped the engagement across three priced tiers (Basic, Standard with accounts, search, and SEO, and Full with Razorpay and Delhivery API) to set a clear boundary and prevent scope creep. Live at silveroaksluxury.com.

Professional Experience

Research Intern, Data Science | *UNSW Sydney, Sydney, NSW | June 2026 – Present*

- Merge MDAD v1.1 event records, Montabone CDOD dust climatology and OpenMARS reanalysis into a single modelling dataset to predict Martian dust storm intensity, supervised by Associate Professor Rohitash Chandra.
- Build reproducible Python pipelines for ingestion, cleaning, temporal alignment and feature engineering across sources of differing spatial and temporal resolution.
- Apply time series and machine learning methods to model storm intensity, evaluating on held-out data with standard regression error metrics and documenting limitations for academic review.

Data Analyst and Web Development Intern | *Nippon Color, Mumbai, India | April 2023 – September 2023*

- Built Power BI dashboards tracking daily traffic, conversion funnels, and revenue, turning raw web analytics into questions the team could act on (where users drop off, what each drop-off costs, which page is leaking), with funnel insights attributed to a 12% increase in online sales.
- Modelled revenue attribution by acquisition channel and product line, and delivered self-serve operational reports stakeholders could open and refresh themselves, removing the analyst as a reporting bottleneck.
- Redesigned the corporate site's information architecture and front-end for a faster, lighter, mobile-first experience, cutting page-load time by 30% and bounce rate by 18%.
- Ran end-to-end testing across devices and browsers before release, covering mobile responsiveness, layout integrity, and first-time user journeys, checking whether attention landed on the actions the business wanted, then logged and resolved the defects found.
- Measured the cleaner post-launch traffic through the BI layer to confirm the changes held in production rather than just in testing.
- Worked inside real Agile sprints with marketing, design, and engineering, owning a delivery thread, aligning deliverables, and resolving blockers to accelerate timelines. Site live at nipponcolor.com.

Data Research and Analyst Intern | *ISchoolConnect, Mumbai, India | December 2021 – June 2022*

- Consolidated and standardised 15,000+ university-program records with Excel Power Query and SQL, improving data governance and lifting data accuracy by 25% to make student search results more reliable.
- Designed web-scraping and Power Automate workflows for data collection, halving collection time and freeing analyst hours for higher-value analysis.
- Automated data validation with Python and pandas scripts, cutting manual cleansing effort by 35% and shortening reporting cycles.
- Built pivot- and lookup-based Excel reporting for recurring stakeholder reports, and facilitated Agile scrum ceremonies for the team.
- Prototyped a Flutter mobile app with a 4-member team, reducing student program-search time by 60% and improving user experience.
- Mentored and trained 4 junior analysts on Excel automation and data-quality best practices, raising team productivity by 20% and reducing onboarding time.

Education

Master of Data Science and Decisions | *UNSW Sydney, Sydney, NSW*
Expected September 2026 | *Concentration: Business Data Science* | *WAM 75.3*

Bachelor of Technology, Computer Engineering | *NMIMS University, Mumbai, India*
2018 – 2024 | *Concentration: Artificial Intelligence* | *CGPA 3.72/4*

Achievements

- Three IEEE conference publications in time series forecasting, predictive maintenance and natural language processing.
 - 1st Prize, Plastic-Free Rivers Hackathon (REVA University and Kyndryl), September 2023: 750 teams across 19 countries.
 - 2nd Place, SAAVISHKAR 2024, NMIMS Computer Engineering final-year project competition, for an adaptive curriculum recommender with a class-level teacher dashboard.
 - "Shut up and take my money" award, Terrible Ideas Hackathon 2025, UNSW Founders.
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Additional Experience and Volunteering

New Year's Eve Volunteer | *City of Sydney* | *December 2024*

- Guided a large, diverse public through one of Sydney's biggest events, giving clear directions and staying calm and approachable in a high-pressure, high-noise setting.
- Answered questions and resolved issues on the spot, adjusting how I explained things to each person so attendees felt looked after rather than processed.
- Worked closely with event staff and local authorities, passing information along accurately and quickly to keep thousands of visitors safe and moving.

FoodHub Volunteer | *Arc @ UNSW* | *May 2026 – Present*

- Supported students using the campus food-relief hub with a warm, non-judgmental manner that kept a sensitive service welcoming and easy to use.
- Kept stock organised and shelves replenished through busy periods so the hub ran smoothly and people were served without long waits.

Waiter (Part-Time) | *Taj Indian Sweets & Restaurant, Sydney* | *November 2024 – November 2025*

- Built quick rapport with a steady flow of customers (50+ interactions per shift), reading what each table needed and keeping service attentive even at peak times.
- Kept front-of-house and kitchen in sync under pressure, communicating orders clearly so dishes went out accurate and on time and tables turned over faster.
- Stayed composed handling several tables at once, and managed POS and payments precisely for smooth, error-free transactions.